Data structures and Algorithms Lab
Assignment 1
Experiment 1: To Demonstrate the Use of Array, Structure, Union and Dynamic Memory Allocation

Instruction: Make a pdf file of the program code in C/C++ and its output.

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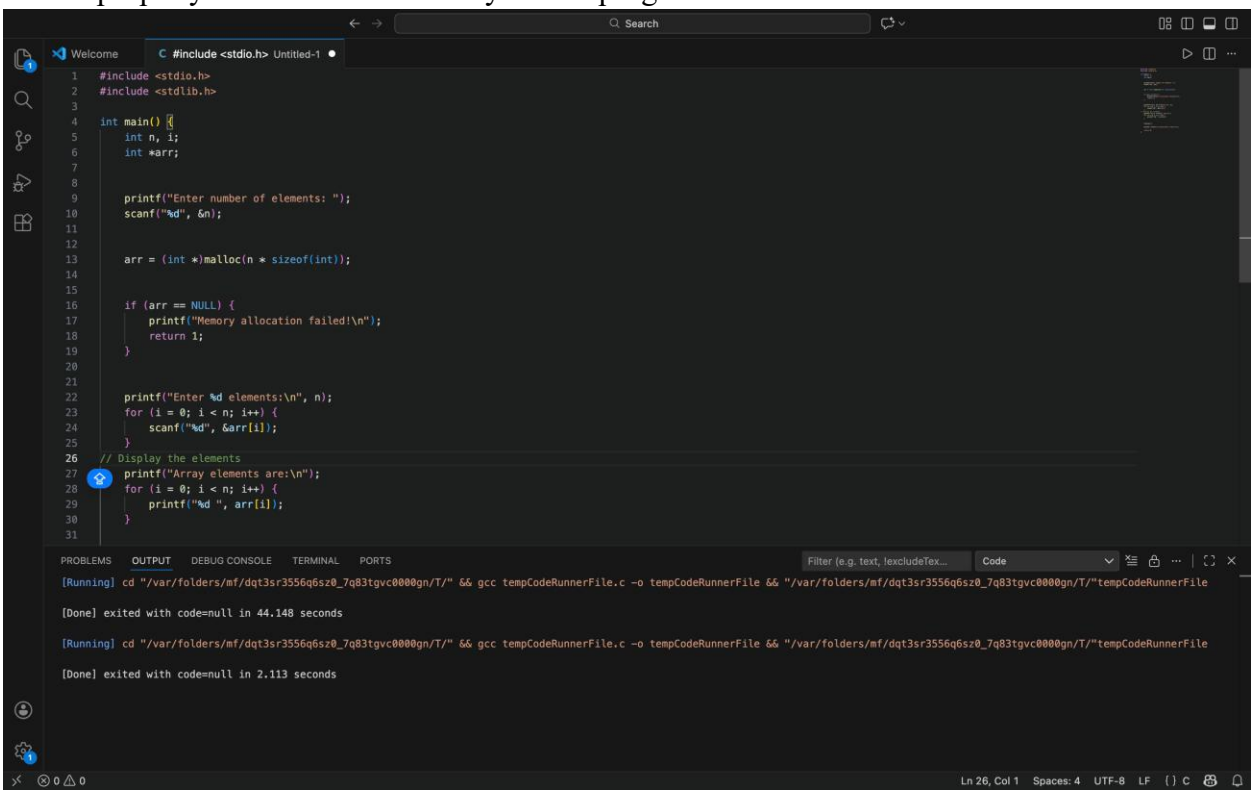
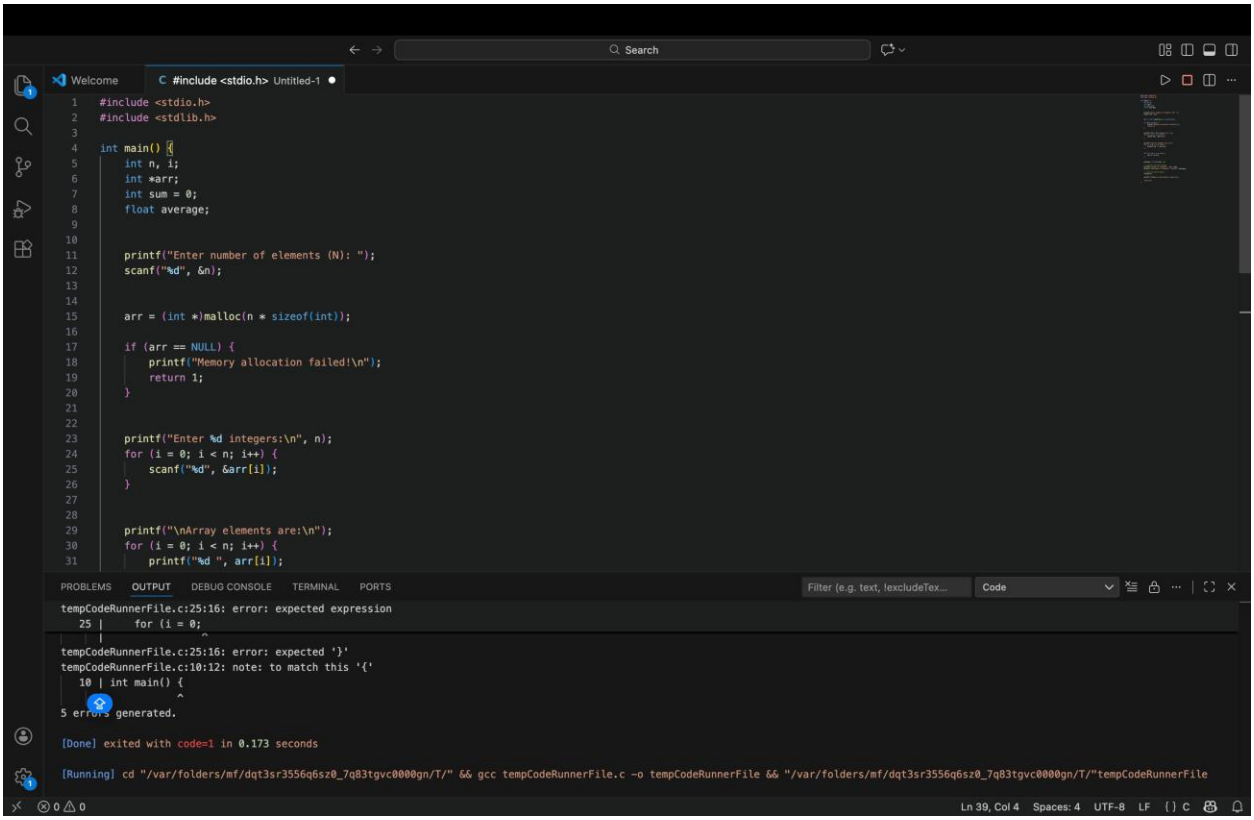
Course: B.Tech. CSE (Sem II)
Subject: DSA Lab
Date: 03 February 2026
Note: Do all the implementations in C programming.

1. Write a C program to dynamically allocate memory for an array, display elements, and properly free allocated memory before program termination.



2. Write a C program to dynamically allocate memory for an array of N integers and perform insertion, display, sum, and average operations.



3. Write a C program to dynamically allocate an array and find the largest and smallest element.

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n, i;
    int *arr;
    int largest, smallest;


    printf("Enter number of elements: ");
    scanf("%d", &n);


    arr = (int *)malloc(n * sizeof(int));
    if (arr == NULL) {
        printf("Memory allocation failed!\n");
        return 1;
    }


    printf("Enter %d integers:\n", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    
    rgest = smallest = arr[0];


    for (i = 1; i < n; i++) {
        if (arr[i] > largest)
```

```
tempCodeRunnerFile.c:10:12: note: to match this '{'
   10 | int main() {
5 errors generated.

[Done] exited with code=1 in 0.173 seconds

[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile
/bin/sh: line 1: 94616 Terminated: 15          "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=143 in 124.196 seconds
```

4. Write a C program to dynamically allocate memory for a 2D matrix and perform matrix addition and subtraction.

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int r, c, i, j;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &r, &c);


    int **A = (int **)malloc(r * sizeof(int *));
    int **B = (int **)malloc(r * sizeof(int *));
    int **Sum = (int **)malloc(r * sizeof(int *));
    int **Diff = (int **)malloc(r * sizeof(int *));

    for (i = 0; i < r; i++) {
        A[i] = (int *)malloc(c * sizeof(int));
        B[i] = (int *)malloc(c * sizeof(int));
        Sum[i] = (int *)malloc(c * sizeof(int));
        Diff[i] = (int *)malloc(c * sizeof(int));
    }

    
    intf("Enter elements of Matrix A:\n");
    or (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            scanf("%d", &A[i][j]);

    printf("Enter elements of Matrix B:\n");
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
```

```
tempCodeRunnerFile.c:10:12: note: to match this '{'
   10 | int main() {
5 errors generated.

[Done] exited with code=1 in 0.173 seconds

[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile
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[Done] exited with code=143 in 124.196 seconds

[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile
```

5. Write a C program using structure and dynamic memory allocation to store and display student details such as roll number, name, and marks.

```c
#include <stdio.h>
#include <stdlib.h>

struct Student {
    int roll;
    char name[50];
    float marks;
};

int main() {
    int n, i;
    struct Student *s;

    printf("Enter number of students: ");
    scanf("%d", &n);


    s = (struct Student *)malloc(n * sizeof(struct Student));
    if (s == NULL) {
        printf("Memory allocation failed!\n");
        return 1;
    }


    for (i = 0; i < n; i++) {
        printf("\nStudent %d\n", i + 1);
        printf("Roll number: ");
        scanf("%d", &s[i].roll);

        printf("Name: ");
        scanf(" %[^\n]", s[i].name);
```

```
/bin/sh: line 1: 94616 Terminated: 15          "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=143 in 124.196 seconds

[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=null in 36.08 seconds

[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=null in 5.628 seconds
```

6. Write a C program using array of structures with dynamic memory allocation to store employee details and find the employee with the highest salary.

```c
#include <stdio.h>
#include <stdlib.h>

struct Employee {
    int id;
    char name[50];
    float salary;
};

int main() {
    int n, i, maxIndex = 0;
    struct Employee *emp;

    printf("Enter number of employees: ");
    scanf("%d", &n);


    emp = (struct Employee *)malloc(n * sizeof(struct Employee));
    if (emp == NULL) {
        printf("Memory allocation failed!\n");
        return 1;
    }


    r (i = 0; i < n; i++) {
        printf("\nEmployee %d\n", i + 1);

        printf("ID: ");
        scanf("%d", &emp[i].id);

        printf("Name: ");
```

```
[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=null in 36.08 seconds

[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=null in 5.628 seconds

[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=null in 1.475 seconds
```

7. Write a menu-driven C program using structure and dynamic memory allocation to add, display, delete records and free memory before exit.

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

struct Student {
    int id;
    char name[50];
    float marks;
};

int main() {
    struct Student *arr = NULL;
    int count = 0, choice;

    do {
        printf("\n--- MENU ---\n");
        printf("1. Add Record\n");
        printf("2. Display Records\n");
        printf("3. Delete Record by ID\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        if (choice == 1) {

            r = realloc(arr, (count + 1) * sizeof(struct Student));
            f (arr == NULL) {
                printf("Memory allocation failed!\n");
                return 1;
            }
```

```
tempCodeRunnerFile.c:11:12: note: to match this '{'
   11 | int main() {
      |            ^
4 errors generated.

[Done] exited with code=1 in 0.055 seconds

[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=null in 0.976 seconds
```

8. Write a C program to demonstrate the use of union by storing integer, float, and character data and explain memory sharing.

```c
#include <stdio.h>

struct MyStruct {
    int i;
    float f;
    char c;
};

union MyUnion {
    int i;
    float f;
    char c;
};

int main() {
    printf("Size of int   : %lu bytes\n", sizeof(int));
    printf("Size of float : %lu bytes\n", sizeof(float));
    printf("Size of char  : %lu bytes\n\n", sizeof(char));

    printf("Size of structure: %lu bytes\n", sizeof(struct MyStruct));
    printf("Size of union    : %lu bytes\n", sizeof(union MyUnion));

    return 0;
}
```

```
[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=null in 0.976 seconds

[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=null in 0.853 seconds

[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=null in 0.855 seconds
```

9. Write a C program to show the difference in memory allocation between structure and union using sizeof().


```c
#include <stdio.h>

struct MyStruct {
    int i;
    float f;
    char c;
};

union MyUnion {
    int i;
    float f;
    char c;
};

int main() {
    printf("Size of int   : %lu bytes\n", sizeof(int));
    printf("Size of float : %lu bytes\n", sizeof(float));
    printf("Size of char  : %lu bytes\n\n", sizeof(char));

    printf("Size of structure: %lu bytes\n", sizeof(struct MyStruct));
    printf("Size of union    : %lu bytes\n", sizeof(union MyUnion));

    return 0;
}
```

```
[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=null in 0.976 seconds

[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=null in 0.853 seconds

[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=null in 0.855 seconds
```


10. Write a C program using structure and dynamic memory allocation to sort student record based on marks.

Search

Welcome | #include <stdio.h> Untitled-1

```c
#include <stdio.h>
#include <stdlib.h>

struct Student {
    int roll;
    char name[50];
    float marks;
};

int main() {
    int n, i, j;
    struct Student *s, temp;

    printf("Enter number of students: ");
    scanf("%d", &n);


    s = (struct Student *)malloc(n * sizeof(struct Student));
    if (s == NULL) {
        printf("Memory allocation failed!\n");
        return 1;
    }


    r (i = 0; i < n; i++) {
        printf("\nStudent %d\n", i + 1);
        printf("Roll number: ");
        scanf("%d", &s[i].roll);

        printf("Name: ");
        scanf(" %[^\n]", s[i].name);
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

Filter (e.g. text, !excludeTex... | Code

```
[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=null in 0.853 seconds

[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=null in 0.855 seconds

[Running] cd "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/" && gcc tempCodeRunnerFile.c -o tempCodeRunnerFile && "/var/folders/mf/dqt3sr3556q6sz0_7q83tgvc0000gn/T/"tempCodeRunnerFile

[Done] exited with code=null in 0.778 seconds
```

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